Chapter Two
Profits

In day to day life all of us are associated with buying-selling and mutual transactions. Someone invests money in industry, produces commodities and sell those commodities to the wholesale traders in the markets. Then the wholesale traders sell those commodities to the retail traders. Finally, the retail traders sell their commodities to the common buyers. In each stage everyone wants to make a profit. But, there may also be loss due to different causes, such as, in the share market, as there is profit, there is also loss due to the fall of market price. Again, we deposit our money in the banks for safety. Bank invests that money in different sectors and makes a profit. Bank also gives profits to the depositors. Therefore, everyone needs to have a clear idea about the investment and the profit. In this chapter, we have discussed profits and losses, especially profits.

At the end of the chapter, the students will be able to –

- explain what profit is;
- explain the rate of simple profit and solve the related problems;
- explain the rate of compound profit and solve the related problems;
- understand and explain the bank's statements.

2.1 Profit and Loss

A businessman adds the shop-rent, transport cost and other related expenditures with the buying price of commodities and fixes the actual cost price, or briefly the cost price. This actual cost price is called investment and this investment is considered to be the buying price for determining profit or loss. The price of selling the commodities is its selling price. If selling price is more than its cost price, there will be a profit. On the other hand, if the selling price is less than the cost price, there will be a loss. Again, if the cost price and the selling price are equal, there will be no profit and no loss. Profit or loss is determined by its cost price.

We can write, Profit = selling price – cost price

Loss = cost price – selling price

From the above relations the cost price or the selling price can be determined.

For comparison, profit or loss is also expressed in percentage.

Example 1. If a shopkeeper buys eggs at Tk. 25 per quad (hali) and sells at Tk. 56 per 2 quads, how much profit will he make ?

Solution : The cost price of 1 quad of eggs = Tk. 25

$\therefore$ The cost price of 2 quads of eggs = Tk. 25×2 = Tk. 50

Again, the selling price of 2 quads of eggs = Tk. 56

Since the selling price is more than the cost price, there will be a profit.

Here, Profit = Tk. $(56 - 50)$ = Tk. 6

In Tk. 50, profit = Tk. 6

In Tk. 1, profit = Tk. $\frac{6}{50}$

In Tk. 100, profit = Tk. $\frac{6 \times 1\cancel{00}^{2}}{\cancel{50}_{1}}$ = Tk. 12

$\therefore$ The profit is 12%

Example 2. A goat is sold at the loss of Tk. 8%. If it were sold at Tk. 800 more, there would be a profit of Tk. 8%. What was the cost price of the goat ?

Solution : If the cost price of the goat was Tk. 100, its selling price would be Tk. $(100 - 8)$, or Tk. 92 at the loss of Tk. 8%.

Again, if the goat was sold at the profit of Tk. 8%, the selling price would be Tk. $(100 + 8)$, or Tk. 108.

$\therefore$ the cost price is more by Tk. $(108 - 92)$, or Tk. 16

If the selling price were Tk. 16 more, the cost price would be Tk. 100

If the selling price ,, ,, 1 ,, ,, ,, Tk. $\frac{100}{16}$

,, ,, ,, ,, ,, 800 ,, ,, ,, Tk. $\frac{100 \times \cancel{800}^{50}}{\cancel{16}_{1}}$

= Tk. 5000

$\therefore$ the cost price of the goat is Tk. 5000.

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Activity : Fill in the blank spaces :			
Cost Price (Tk.)	**Selling price (Tk.)**	**Profit/Loss**	**Percentage of Profit/Loss**
600	660	Profit Tk. 60	Profit 10%
600	552	Loss Tk. 48	Loss 8%
	583	Profit Tk. 33	
856		Loss Tk. 107	
		Profit Tk. 64	Profit 8%

2.2 Profit :

Farida Begum decided to deposit her savings in a bank. She deposited Tk. 10,000 in a bank. After one year she went to the bank to take a bank statement. She noticed that her deposited amount of money had been increased by Tk. 700 and her balance became Tk. 10,700. How was the amount of money of Farida Begum increased by Tk. 700?

When money is deposited in a bank, the bank invests that money as loans in different sectors, such as business, house-building etc. and gets profit from those sectors. From that profit, the bank gives some money to the depositor. This money is the profit of the depositor. The money which was first deposited in the bank was her principal amount. To deposit or give money as a loan and to take money from anyone as a loan is accomplished by a process. This process relates to the principal, the rate of profit and time and profit.

We observe :

Rate of Profit : A profit of Tk. 100 in 1 year is called the rate of profit or percentage of profit per annum.

Period of time : The time for which a profit is calculated, is called the period of time.

Simple profit : The profit which is accounted each year only on the primary or initial principal, is called the simple profit. Usually, profit means the simple profit. In this chapter we shall use the following algebraic symbols :

Principal = P Rate of profit = r (rate of interest) Time = n Profit = I Profit-principal = A (total amount)	Profit-Principal = Principal + Profit i.e. ; $A = P + I$ which gives $P = A - I$ $I = A - p$

2.3 Problems related to profit

If any three of the four data namely principal, rate of profit, time and profit are known, the remaining one can be found. It is discussed as follows :

(a) Determination of profit :

Example 3. Mr. Ramiz deposited Tk. 5,000 in a bank and decided not to withdraw any amount from the bank till next 6 years.The annual profit given by the bank is 10%. How much profit will he get after 6 years ? What will be the profit-principal ?

Solution : The profit of Tk. 100 in 1 year is Tk. 10

$$\text{,, Tk. } 1 \text{ ,, } 1 \text{ ,, ,, Tk. } \frac{10}{100}$$

$$\text{,, Tk. } 5{,}000 \text{ ,, } 1 \text{ ,, ,, Tk. } \frac{10\times 5{,}000}{100}$$

$$\text{,, Tk. } 5{,}000 \text{ ,, } 6 \text{ ,, ,, Tk. } \frac{10\times \cancel{5000}^{50}\times 6}{\cancel{100}_{1}} = \text{Tk. } 3{,}000$$

$\therefore$ Profit-principal = principal + profit

= Tk. (5,000 + 3,000)

= Tk. 8,000.

$\therefore$ Profit is Tk. 3,000 and Profit-principal is Tk. 8,000.

We observe : Profit of Tk. 5,000 in 6 years at the percentage of Tk. 10 per annum = Tk. $\left(5{,}000\times\frac{10}{100}\times 6\right)$.

Formula : Profit = Principal × Rate of Profit × time, $I=Prn$
Profit-Principal = Principal + Profit, $A=P+I=P+Prn=P(1+rn)$

Alternative solution to example 3 :

We know, $I=Prn$, i.e., Profit = Principal × rate of profit × time

$\therefore$ Profit = Tk. $\left(5{,}000\times\frac{10}{100}\times 6\right)$

= Tk. 3,000

$\therefore$ Profit-Principal = Principal + Profit

= Tk. (5,000 + 3,000) or Tk. 8,000

$\therefore$ Profit is Tk. 3,000 and Profit-Principal is Tk. 8,000.

(b) Determination of Principal :

Example 4. If the rate of profit is $8\frac{1}{2}\%$ per annum, what amount will make profit Tk. 2,550 in 6 years ?

Solution : Rate of profit $= 8\frac{1}{2}\%$ or $\frac{17}{2}\%$

We know, $I = Prn$ or, $P = \frac{I}{rn}$

$\therefore$ Principal $= \frac{\text{profit}}{\text{rate of profit} \times \text{time}}$

$$= \text{Tk. } \frac{2{,}550}{\frac{17}{2 \times 100} \times 6} = \text{Tk. } \frac{\overset{50}{\cancel{\overset{150}{\cancel{2550}}}} \times \overset{1}{\cancel{2}} \times 100}{\underset{1}{\cancel{17}} \times \underset{1}{\cancel{\underset{3}{\cancel{6}}}}}$$

$= \text{Tk. } (50 \times 100) = \text{Tk. } 5{,}000.$

$\therefore$ Principal is Tk. 5,000.

Where,

P = Principal = Required

I = Profit = Tk. 2,550

r = Rate of Profit

$= 8\frac{1}{2}\%$ or $\frac{17}{2}\%$

n = Time = 6 years

(c) Determination of the rate of profit:

Example 5. What is the rate of profit by which the profit of Tk. 3,000 will be Tk. 1,500 in 5 years ?

Solution : We know, $I = Prn$ or, $r = \frac{I}{Pn}$

$\therefore$ rate of profit $= \frac{\text{profit}}{\text{principal} \times \text{time}}$

$$\frac{\overset{1}{\cancel{1500}}}{\underset{2}{\cancel{3000}} \times 5}$$

$$= \frac{1}{10} = \frac{1 \times 100}{10 \times 100}$$

$= 10\%$

$\therefore$ The rate of profit is 10%

Where,

P = Principal = Tk. 3,000

I = Profit = Tk. 1,500

r = Rate of Profit = Required

n = Time = 6 years

Example 6. The profit-principal of some principal is Tk. 5,500 in 3 years. If the profit is $\frac{3}{8}$ part of the principal, find the principal and the rate of profit ?

Solution : We know, principal + profit = profit-principal

i.e. $P+I=A$

or, $P+\frac{3}{8}P=A$

or, $\left(1+\frac{3}{8}\right)\times \text{P} = 5{,}500$

or, $\frac{11}{8}\times \text{P} = 5{,}500$

$\therefore$ P $=$ Tk. $\frac{\overset{500}{\cancel{5500}}\times 8}{\underset{1}{\cancel{11}}}$

= Tk. 4,000.

Where,

P = Principal

$\therefore$ Profit = profit-principal – principal

= Tk. (5,500 – 4,000) or, Tk. 1,500

Again, we know, $I = Prn$

or, $r=\frac{I}{Pn}$

$=$ Tk. $\frac{1500}{4000\times 3}$

$=\frac{\overset{25}{\cancel{500}}\;\cancel{1500}\times\cancel{100}}{\cancel{4000}\;\underset{2}{\cancel{40}}\times\cancel{3}}\%$ or, $\frac{25}{2}\%$ or, $12\frac{1}{2}\%$

Where,

P = Principal = Tk. 4,000

I = Profit = Tk. 1,500

r = Rate of Profit = Required

n = Time = 3 years

$\therefore$ Principal is Tk. 4,000 and rate of profit is $12\frac{1}{2}\%$

(D) Determination of time :

Example 7. In how many years will the profit of Tk. 10,000 be Tk. 4,800 at the rate of 12% profit?

Solution : We know, $I = Prn$

or, $n=\frac{I}{Pr}$

where profit I = Tk. 4,800, Principal P = Tk. 10,000

rate of profit r =12%, time n = ?

$$\therefore \text{ Time } n = \frac{\text{profit}}{\text{principal} \times \text{rate of profit}}$$

$$\therefore \text{ Time } = \frac{4{,}800}{10{,}000 \times \frac{12}{100}} \text{ years}$$

$$= \frac{\overset{4}{\cancel{48}}\cancel{4800} \times \cancel{100}^{1}}{\cancel{10000}\,\underset{1}{\cancel{100}} \times \underset{1}{\cancel{12}}} \text{ years}$$

$$= 4 \text{ years}$$

$\therefore$ Time is 4 years.

Exercise 2.1

1. On selling a commodity, the profit of a wholesale seller is 20% and the profit of retail seller is 20%. If the retail selling price of the commodity is Tk. 576, what is the cost price of the wholesaler?
2. A shopkeeper sold some amount of pulses for Tk. 2,375 at the loss of Tk. 5%. What would be the selling price of pulses to make a profit of Tk. 6% ?
3. An equal number of bananas is bought at 10 and 15 pieces per Tk. 30 and all the bananas are sold at 12 pieces per Tk. 30. What will be the percentage of profit or loss?
4. What will be the profit for Tk. 2,000 in 5 years if the percentage of profit is Tk. 10.50 per annum?
5. How much less will be the profit of Tk. 3,000 in 3 years if the percentage of profit per annum is decreased from Tk. 10 to Tk. 8 ?
6. What is the percentage of profit per annum by which Tk. 13,000 will be Tk. 18,850 as profit-principal in 5 years ?
7. For what percentage of profit per annum, some principal will be double in profit-principal in 8 years ?
8. How much money will become Tk. 10,200 as profit-principal in 4 years at the same rate of profit at which Tk. 6,500 becomes Tk. 8,840 as the profit-principal in 4 years ?

9. Mr. Riaz deposited some money in a bank and got the profit of Tk. 4,760 after 4 years. If the percentage of profit of the bank is Tk. 8.50 per annum, what amount of money did he deposit in the bank ?

10. What amount of money will become Tk. 2,050 as profit-principal in 4 years, at the same rate of profit at which some principal becomes double as profit-principal in 6 years ?

11. For how much money will the profit at the rate of Tk. 5 per annum in 2 years 6 months be same as that ofTk. 500 at the rate ofTk. 6 per annum in 4 years?

12. Due to increase in the rate of profit from 8% to 10%, the income of Tisha Marma was increased by Tk. 128 in 4 years. How much was her principal?

13. Some principal becomes Tk. 1,578 as profit-principal in 3 years and Tk. 1,830 as profit-principal in 5 years. Find the principal and the rate of profit.

14. Tk. 3,000 at the rate of 10% profit and Tk. 2,000 at the rate of 8% profit are invested. What will be the average percentage of profit on the total sum of the principals?

15. Rodrick Gomage borrowed Tk. 10,000 for 3 years and Tk. 15,000 for 4 years from a bank and paid Tk. 9,900 in total as a profit. In both cases if the rate of profit is same, find the rate of profit.

16. Some principal becomes its double as profit-principal in 6 years at the same percentage of profit. In how many years will it be thrice of it as profit-principal at the same percentage of profit ?

17. The profit-principal for a certain period of time is Tk. 5,600 and the profit is $\frac{2}{5}$ of the principal. If the percentage of profit is Tk. 8, find the principal and time.

18. After having the pension, Mr. Jamil bought pension savings certificates of Tk. 10 lac for five years term on the basis of having the profit in three months' interval. If the percentage of profit is 12% per annum, what amount of profit will he get at the first installment, that is, after first three months ?

19. A fruit seller bought some bananas at the cost of Tk. 36 for 12 pieces from Jessore and Tk. 36 for 18 pieces from Kustia. He bought equal pieces of bananas both from Jessore and Kustia. His salesman sold the bananas at Tk. 36 for 15 pieces.

A. What was the cost price of 100 pieces from Jessore?
B. If the salesman sold all bananas, how much would be profit or loss?
C. If the fruit seller wants to make 25% profit, what would be the selling price for 4 pcs of banana?

20. A principal is turned to an amount in 3 years of Tk. 28,000 and in 5 years of Tk. 30,000 at simple interest.
   A. Using the symbols in details, write the formula for Principal.
   B. Find out the rate of interest or profit.
   C. How much principal should be deposited to get the amount of Tk. 48,000 at the same rate of interest ?

2.4 Compound Profit

In the case of compound profit, the profit of any amount of principal is added to the principal at the end of each year and the total sum is considered as the new principal. If a depositor deposits Tk. 1,000 in a bank and the bank gives him the profit at the rate of 12%, the depositor will get profit on Tk. 1,000 at the end of one year.

$$\begin{aligned} 12\% \text{ of Tk. } 1{,}000 &= \text{Tk. } 1{,}000 \times \frac{12}{100} \\ &= \text{Tk. } 120. \end{aligned}$$

Then, for the 2nd year, his principal will be Tk. (1000 + 120) = Tk. 1,120, which is his compound principal. Again, 12% profit will be given on Tk. 1,120 at the end of 2nd year.

$$\begin{aligned} 12\% \text{ of Tk. } 1120 &= \overset{224}{\cancel{1120}} \times \frac{\cancel{12}}{\underset{\underset{5}{\cancel{25}}}{\cancel{100}}} \\ &= \text{Tk. } \frac{672}{5} \\ &= \text{Tk. } 134.40. \end{aligned}$$

So, for the 3rd year the compound principal of the depositor will be Tk. (1120+ 134.40) = Tk. 1254.40

In this way, the principal of the depositor at the end of each year will go on to be increased. This increased principal is called the compound principal and the profit which is given on the increased principal, is called the compound profit. Terms for giving profit may be for three months or six months or may be less than those periods.

Formation of the formulae for compound principal and compound profit :

Let the initial principal be P and the percentage of compound profit be r per annum. Then, at the end of 1st year, compound principal = principal + profit

$$=P+P\times r$$
$$=P(1+r)$$

Again, at the end of 2nd year, Compound principal = compound principal of the 1st year + profit

$$= P(1+r) + P(1+r)\times r$$
$$= P(1+r)(1+r)$$
$$= P(1+r)^2$$

At the end of 3rd year, Compound principal = compound principal of the 2nd year + profit

$$= P(1+r)^2 + P(1+r)^2\times r$$
$$= P(1+r)^2(1+r)$$
$$= P(1+r)^3$$

We observe : In the compound principal
at the end of 1st year, index of $(1+r)$ is 1
at the end of 2nd year, index of $(1+r)$ is 2
at the end of 3rd year, index of $(1+r)$ is 3

$\therefore$ at the end of nth year, index of $(1+r)$ will be n .

$\therefore$ if C be the compound principal at the end of n years, then $P(1+r)^n$.

Again, compound profit = Compound principal - Initial principal= $P(1+r)^n - P$

Formula : Compound principal $C = P(1+r)^n$
Compound profit $= C - P = P(1+r)^n - P$

Now, we apply the formula for compound principal in case where the initial principal of Tk. 1,000 and profit of 12% were taken at the beginning of the discussion about compound principal :

At the end of 1st year, compound principal $= P(1+r)$

$$= \text{Tk. } 1{,}000 \times \left(1+\frac{12}{100}\right)$$

$$= \text{Tk. } 1{,}000 \times (1+0.12)$$

$$= \text{Tk. } 1{,}000 \times 1.12$$

$$= \text{Tk. } 1{,}120$$

At the end of 2nd year, compound principal $= P(1+r)^2$

$$= \text{Tk. } 1{,}000 \times \left(1+\frac{12}{100}\right)^2$$

$$= \text{Tk. } 1{,}000 \times (1+0.12)^2$$

$$= \text{Tk. } 1{,}000 \times (1.12)^2$$

$$= \text{Tk. } 1{,}000 \times 1.2544$$

$$= \text{Tk. } 1{,}254.40$$

At the end of 3rd year, compound principal $= P(1+r)^3$

$$= \text{Tk. } 1{,}000 \times \left(1+\frac{12}{100}\right)^3$$

$$= \text{Tk. } 1{,}000 \;\; (1+0.12)^3$$

$$= \text{Tk. } 1{,}000 \times (1.12)^3$$

$$= \text{Tk. } 1{,}000 \times 1.404928$$

$$= \text{Tk. } 1{,}404.93 \text{ (approx.)}$$

Example 1. Find the compound principal of Tk. 62,500 in 3 years at the profit of Tk. 8 percent per annum.

Solution : We know, compound principal. $C = P\,(1+r)^n$.

Given, initial principal P = Tk. 62,500

percentage of profit r = 8%

and time n = 3 years

$$\therefore\ C = 62{,}500 \times \text{Tk.}\left(1+\frac{\overset{2}{\cancel{8}}}{\underset{25}{\cancel{100}}}\right)^3 = \text{Tk.}\,62{,}500 \times \left(\frac{27}{25}\right)^3$$

$$= \text{Tk. } 62{,}500 \times (1.08)^3$$

$$= \text{Tk. } 78{,}732$$

$\therefore$ Compound principal is Tk. 78,732

Example 2. Find the compound profit of TK. 5000 at the profit of 10.50% per annum in 2 years.

Solution : To find out the compound profit, at first we find the compound principal.

We know,

compound principal $C = P(1+r)^n$, where principal P = Tk. 5,000

percentage of profit $r = 10.50\% = \frac{21}{200}$

and time n = 2 years.

$$\therefore\ C = P(1+r)^2$$

$$= \text{Tk. } 5{,}000 \times \left(1+\frac{21}{200}\right)^2$$

$$= \text{Tk. } 5{,}000 \times \left(\frac{221}{200}\right)^2$$

$$= \text{Tk. } \overset{\overset{1}{\cancel{25}}}{\cancel{5000}} \times \frac{221}{\underset{1}{\cancel{200}}} \times \frac{221}{\underset{8}{\cancel{200}}}$$

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$= \text{Tk. } \frac{48,841}{8}$ or Tk. 6,105.13 (approx.)

$\therefore$ Compound profit $= C - P = P\,(1+r)^2 - P$

$= \text{Tk. } (6,105.13 - 5,000)$

$= \text{Tk. } 1,105.13$ (approx.)

Example 3. A flat owners welfare association deposited the surplus money of Tk. 2,00,000 from its service charges in a bank in the fixed deposit scheme on the basis of compound profit for six months' interval. If the percentage of profit is Tk. 12, how much profit will be credited to the account of the association ? What will be the compound principal after one year ?

Solution : Given, principal $P = \text{Tk. } 2,00,000$

rate of profit $r = 12\%$, time $n = 6 \text{ months} = \frac{1}{2}$ year

$\therefore$ profit $= \text{Tk. } I = Prn$

$$= \text{Tk. } \overset{2,000}{\cancel{2,00,000}} \times \frac{\cancel{12}^{\,6}}{\cancel{100}_{\,1}} \times \frac{1}{\cancel{2}_{\,1}}$$

$= \text{Tk. } 12,000$

$\therefore$ profit after 6 months will be Tk. 12,000

After 6 months compound principal=Tk. (2,00,000+12,000) = Tk. 2,12,000

Again, profit-principal after 6 months $= \text{Tk. } 2,12,000\,(1+\frac{12}{100}\times\frac{1}{2})$

$= \text{Tk. } 2,12,000 \times 1.06$

$= \text{Tk. } 2,24,720$

$\therefore$ Compound principal after 1 year will be Tk. 2,24,720.

Example 4. The present population of a city is 80 lac.What will be the population of the city after 3 years if the growth rate of population of that city is 30 per thousand?

Solution : present population of the city is $P = 80{,}00{,}000$

growth rate of population, $r = \dfrac{30}{1000} \times 100\% = 3\%$

time $n = 3$ years.

Here, in the case of growth of population, formula for compound principal is applicable.

$\therefore$ $C = P\,(1+r)^n$

$$= 80{,}00{,}000 \times \left(1 + \frac{3}{100}\right)^3$$

$$= 80{,}00{,}000 \times \frac{103}{100} \times \frac{103}{100} \times \frac{103}{100}$$

$$= 8 \times 103 \times 103 \times 103$$

$$= 87{,}41{,}816$$

$\therefore$ After 3 years, population of the city will be 87,41,816.

Example 5. To meet an urgent family need, Monowara Begum takes a loan of taka 'x' at the rate of 6% interest and taka 'y' at 4% interest. She takes in total taka 56,000 as a loan and pays Tk. 2,840 as interest.

A. What will be annual interest if 5% interest is imposed on total loan?

B. Find out the value of x and y.

C. How much interest will be paid by Monowara Begum if 5% compound interest is imposed for 2 years ?

Solution :

A. Total amount of loan P = Tk 56,000

Rate of interest, r = 5%

Time, n = 1 year

$\therefore$ Interest I $= Pnr$

$= \text{Tk.}\ (56000 \times 1 \times \frac{5}{100})$

$= \text{Tk.}\ 2800$

B. The annual Interest on Tk. x at 6% = Tk $(x\times1\times\frac{6}{100})$

$= \text{Tk } \frac{6x}{100}$

Again, The annual Interest on Tk 'y' at 4% = Tk $(y\times1\times\frac{4}{100})$

$= \text{Tk } \frac{4y}{100}$

Now, according to the information of the stimulus,

$x + y = 56{,}000$ ......(i)

And $\frac{6x}{100} + \frac{4y}{100} = 2{,}840$

Or, $6x+4y = 2{,}84{,}000$

Or, $3x+2y = 1{,}42{,}000$ ......(ii)

Now multiply equation (i) by 3 and subtract equation (ii)

$$\begin{aligned} 3x+3y &= 168000 \\ \underline{3x+2y} &\underline{= 142000} \\ y &= 26000 \end{aligned}$$

Now, putting the value of 'y' in equation (i)

We get, $x = 30{,}000$

$\therefore$ x = Tk. 30,000 and y = Tk. 26,000

C. Total amount of loan P = Tk. 56,000

Rate of Interest r = 5%

Time n = 2 years.

Now, for compound rate of Interest, the amount of loan = $P(1+r)^n$

$\therefore$ After 2 years, the amount of loan $= \text{Tk. } 56{,}000\,(1+\frac{5}{100})^2$

$= \text{Tk. } 56{,}000 \text{ x } (1.05)^2$

$= \text{Tk. } 61{,}740$

$\therefore$ The interest to be paid by Monowara $= \text{Tk. } (61{,}740\text{-}5{,}600)$

$= \text{Tk. } 5{,}740$

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Exercise 2.2

1. What is the simple profit of Tk. 1,200 in 4 years at the rate of simple profit of 10% per annum?

   a. Tk. 120 b. Tk. 240 c. Tk. 360 d. Tk. 480

2. If the rate of profit is 10% per annum, find the compound principal ofTk. 8000 in 3 years.

3. What will be the difference of simple profit and compound profit of Tk. 5,000 in 3 years if the rate of profit is Tk. 10 percent per annum?

4. What was the principal if the compound principal of any amount of principal at the end of one year is Tk. 6,500 and at the end of two years is Tk. 6,760 at the same rate of profit ?

5. If the rate of compound profit is Tk. 8.50 percent per annum, find the compound principal and compound profit ofTk. 10000 in 2 years.

6. Present population of a city is 64 lac. What will be the population of the city after 2 years if growth rate of population of the city is 25 per thousand ?

7. A person borrows Tk. 5,000 from a lending organization at the rate of 8% compound profit. At the end of every year he pays off Tk.2,000. How much more money will he have as loan after paying off the 2nd installment ?

Sample Questions

Multiple Choice Questions:

1. Which one of the following is 8% of Tk. 1050 ?

   a. Tk. 80 b. Tk. 82 c. Tk. 84 d. Tk. 86

2. The cost price of something is 5 pieces at Tk. 1 and selling price is 4 pieces at Tk. 1. What will be the percentage of profit or loss?

   A. Profit 25% B. Loss 25% C. Profit 20% D. Loss 20%

3. At a 10% interest rate, a 3-year compound interest on Tk. 200 is Tk. 66. Which of the following is true?
   i. Simple interest = Tk. 60
   ii. Compound amount = Tk. 266
   iii. Difference between compound and simple interest = Tk. 6

   Which is correct?
   (a) i and ii (b) i and iii (c) ii and iii (d) i, ii, and iii

   Mr.Jamil deposited Tk. 2,000 in a bank at the rate of profit 10% per annum:

   Answer the following 4 & 5 using stem

4. In simple profit, what will be the profit-principal at the end of 2nd year?
   a. Tk. 2,400 b. Tk. 2,420 c. Tk. 2,440 d. Tk. 2,450
5. What will be the compound principal at the end of 1st year ?
   a. Tk. 2,050 b. Tk. 2,100 c. Tk. 21,500 d. Tk. 2,200

Creative Question:

6. At the same rate of compound profit a principal will amount to Tk. 19,500 after 1 year and Tk. 20,280 after 2 years.
   A. Write down the formula for profit.
   B. Find out the principal.
   C. Find the difference between simple profit and compound profit after 3 years at the same rate for the principal.

Short Answer Questions:

7. (a) At a 5% interest rate, in how many years will Tk. 500 yield Tk. 250 in interest?
   (b) If a product is sold at Tk. 1344 with a 12% profit, what is the cost price?
   (c) At what annual interest rate will Tk. 2500 earn Tk. 600 in 3 years?
   (d) If Tk. 1000 is deposited in a bank at 12% compound interest, what will be the compound amount in 2 years?
   (e) The current population of a city is 30 lakh, and the population is increasing at a rate of 30 per 1000 annually. What will the population be after 1 year?